



Basic Original Report

Evaluation of near-miss and adverse events in radiation oncology using a comprehensive causal factor taxonomy



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Abstract

Purpose: Incident learning systems (ILSs) are a popular strategy for improving safety in radiation oncology (RO) clinics, but few reports focus on the causes of errors in RO. The goal of this study was to test a causal factor taxonomy developed in 2012 by the American Association of Physicists in Medicine and adopted for use in the RO: Incident Learning System (RO-ILS).

Methods and materials: Three hundred event reports were randomly selected from an institutional ILS database and Safety in Radiation Oncology (SAFRON), an international ILS. The reports were split into 3 groups of 100 events each: low-risk institutional, high-risk institutional, and SAFRON. Three raters retrospectively analyzed each event for contributing factors using the American Association of Physicists in Medicine taxonomy.

Results: No events were described by a single causal factor (median, 7). The causal factor taxonomy was found to be applicable for all events, but 4 causal factors were not described in the taxonomy: linear accelerator failure ($n = 3$), hardware/equipment failure ($n = 2$), failure to follow through with a quality improvement intervention ($n = 1$), and workflow documentation was misleading ($n = 1$). The most common causal factor categories contributing to events were similar in all event types. The most common specific causal factor to contribute to events was a “slip causing physical error.” Poor human factors engineering was the only causal factor found to contribute more frequently to high-risk institutional versus low-risk institutional events.

Conclusions: The taxonomy in the study was found to be applicable for all events and may be useful in root cause analyses and future studies. Communication and human behaviors were the most common errors affecting all types of events. Poor human factors engineering was found to specifically contribute to high-risk more than low-risk institutional events, and may represent a strategy for reducing errors in all types of events.

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Conflicts of interest: None.

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Introduction

Efforts to improve patient safety in radiation oncology (RO) have accelerated in recent years. One strategy for improving safety in RO clinics is to implement an incident learning system (ILS).¹⁻³ Incident learning refers to the process of reporting a patient safety near-miss or adverse event, analyzing it in detail, and developing interventions to prevent it from happening again.^{1,4} Numerous prior publications have recommended the use of ILS in RO clinics,^{1,5-8} and ILS has now become a requirement for practice accreditation through the American Society for Radiation Oncology (ASTRO; <https://www.astro.org/Accreditation.aspx>). There have also been efforts to implement multicenter ILS to more broadly address patient safety in RO, such as the Radiation Oncology Incident Learning System (RO-ILS) from ASTRO and the American Association for Physicists in Medicine (AAPM)⁹ and the Radiotherapy Incident Reporting and Analysis System from the Center for Assessment in the Radiological Sciences patient safety organization (www.cars-psy.org).

With increased reporting into ILS, there have been efforts to develop standardized operational frameworks, metrics, and terminologies to learn from errors in RO,^{1,10-14} and prior works have described many features of errors in radiation therapy, such as where they originate in the RO workflow,¹⁵ which clinical features predict event reporting,¹⁶ and the frequencies at which different RO team members report events.^{17,18}

Although there is a substantial and growing literature on incident learning in the RO context, few reports specifically focus on causes of errors in RO.¹⁹⁻²¹ This is arguably the most important aspect of incident learning, because understanding the underlying causes may lead to the prevention or mitigation of errors. A thorough understanding of factors contributing to errors also supports improvement of radiation therapy equipment and practices through strategies such as human factors engineering (HFE). This requires a deeper analysis of the causal factors that drive error beyond reductionist assignments such as “human error.”²² The reports published in the RO literature thus far do not present a deep analysis of this type and are highly variable in their use of causal factors. In 2012, a standardized taxonomy of potential causes of RO errors was proposed by an AAPM workgroup,¹ and this system was adopted for use in the RO-ILS system. There has been no report to date that systematically tests or validates this causal factor schema in the RO context. The goal of this study therefore was to systematically test the causal factor taxonomy to better understand the causes of patient safety events in RO through a retrospective analysis of causal factors leading to near-miss and adverse events in both a departmental ILS and multicenter ILS.

Methods

Three hundred event reports were randomly selected for analysis from our institutional ILS, entered between

February 2011 and September 2015, and from Safety in Radiation Oncology (SAFRON), an international RO incident reporting system maintained by the International Atomic Energy Agency (<https://rpop.iaea.org/SAFRON>). The institutional incident reports were events that did not affect the patient but were deemed to be potentially important safety indicators. The reports were split into 3 groups: low-risk institutional, high-risk institutional, and SAFRON. The low- and high-risk institutional groups contained 100 reports each with low (0-2) and high (3-4) near-miss risk index (NMRI) scores, respectively. The NMRI has been described previously.¹⁴ The third group (eg, SAFRON) contained 100 reports from SAFRON. An attempt was made to randomly select reports from SAFRON, but the final dataset was drawn largely from more recent reports because earlier reports often had insufficient information required for causal factors analysis.

Each event was classified as a near-miss, an incident, or “other,” consistent with the definitions used in the current version of RO-ILS (circa 2017). A near-miss is defined as a safety event that did not reach the patient. An incident is defined as an event that reached the patient with or without harm. Incidents include events in which radiation dose was not delivered as intended and events not involving radiation dose (eg, collision, fall). For instance, an error leading to a repeat computed tomography (CT) simulation scan would be considered an incident in this analysis. Events classified as “other” are of 2 types: unsafe conditions (eg, conditions that increase the probability of a safety event) or operational/process improvements (eg, nonsafety event). Examples of events classified as “other” include a report concerning an unclear policy on isocenter localization procedures (considered an “unsafe condition”) or a problem scheduling a patient’s initial treatment because of unclear assignment of duties (considered a “process improvement”).

Three raters (MS, RF, OG) were trained in the interpretation of the causal factor table and causal factors were scored by consensus. The raters retrospectively analyzed each event for contributing factors using a custom data collection tool created using REDCap.²³ The data collection tool allowed each rater to enter the event date, index number, NMRI, and to select potential factors contributing to the event. The potential contributing factors used in this study were identical to those proposed in the AAPM consensus report on incident learning (Appendix D¹). This causal factor taxonomy was created collaboratively with input from representatives from radiation therapy professional societies (eg, ASTRO, American College of Radiology) and a core developer of Radiation Oncology Safety Information System and SAFRON. The taxonomy was specifically designed to be robust and specific to radiation therapy, easy to use, and mappable to other taxonomies. This specific taxonomy was chosen because the same 19 categories were used in the RO-ILS system. Of note, the causal factors tables in RO-ILS were updated and modified in October 2016 after

completion of data collection. The categories discussed and presented here can still be found in the new system, albeit with slightly different wording and ordering. The taxonomy is organized into 7 top-level categories (6 named plus “other”) (1): organizational management, (2) technical, (3) human behavior involving staff, (4) patient-focused circumstances, (5) external factors (beyond facility control), (6) procedural issues, and (7) other. The taxonomy is further organized into 19 subcategories. This includes “human behavior involving staff” and other, which have no subcategories and are counted as both a top-level and subcategory in this analysis. In total, these include 93 specific potential causal factors that may contribute to errors in RO. For example, the category of “human behavior involving staff” includes the following 6 casual factors (1): acting outside one’s scope of practice, (2) slip causing physical error, (3) poor judgment, (4) language and comprehension issues, (5) intentional rules violations, and (6) negligence. Each event was scored for

contributing factors by consensus using all available information in the submitted incident report.

The proportion of events affected by each of the 93 potential contributing factors was calculated for the low-risk, high-risk, and SAFRON groups. Proportions were compared using χ^2 tests in Microsoft Excel (v14.0.7177.500). The institutional review board approved this study.

Results

Most events in the low-risk institutional group were classified as “other” (71%), whereas most events in the high-risk institutional group were classified as near misses (57%) and the majority of events in the SAFRON group were classified as incidents (85%) (Fig 1A). The proportion of events in the institutional groups classified

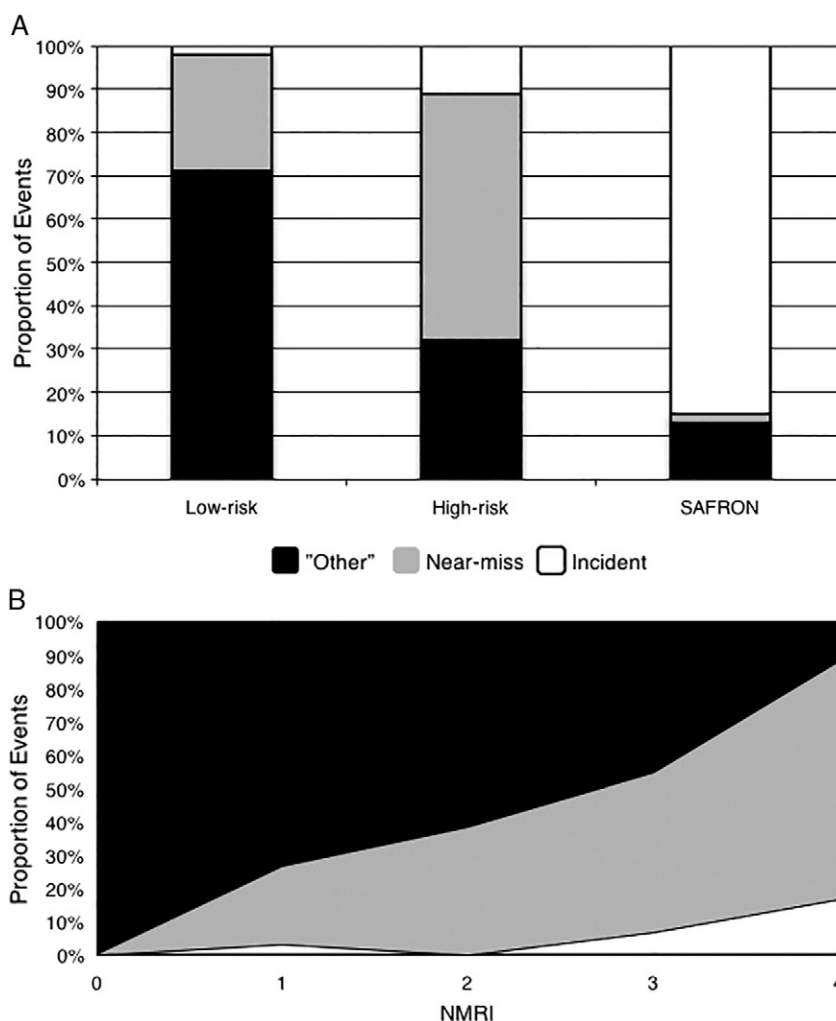


Figure 1 (A) Proportion of events in each group categorized as “other” (eg, “unsafe conditions” or “operational/process improvements,” black), “near-miss” (gray), or “incident” (white). (B) Proportion of events with each near-miss risk index (NMRI) score (0–4) categorized as “other” (black), “near-miss” (gray), or “incident” (white). SAFRON, Safety in Radiation Oncology.

Table 1 Examples of events with specific causal factors in the most common categories

Event narrative	Specific causal factors
Example 1: "Prescription specified to treat the patient 2 fractions per week. The therapists treating the patient failed to notice the specification of 2 fractions per week and scheduled the patient for daily treatment."	• Human slip causing physical error • Medical records incorrect • Loss of attention
Example 2: "The patient had been treated previously to T6-T8, and we wanted to treat T10-T12. The plan was to set up to the old tattoo and shift 6.5 cm inferiorly. The shift did not happen and the field wound up overlapping the previous fields. Changed procedures to require cone beam CT for localization of spine fields."	• Human slip causing physical error • Policy inadequate • Poor judgment • Loss of attention
Example 3: "Physician decided to add photon boost the day after the patient finished initial prescription. Therapists were not informed and there was no note in the chart. The boost prescription was also not signed, and this led to rushed work."	• Poor, incomplete, unclear, or missing communication • Slip causing physical error • Poor judgment • Lack of information • Missing information
Example 4: "New field names created causing confusion with therapist. The name 'iR30A' could mean inferior instead of initial. We initially assumed this is what the dosimetrist intended. It was later discovered that the treatment planning system does not allow beams to export that have the same first 4 letters, so the 'i' was used as a placeholder for export to the record-and-verify system."	• Poor, incomplete, unclear, or missing communication • Relevant policy nonexistent • Unclear communication • Poor human factors engineering • Interoperability problem • Failure to request needed information • Loss of attention
Example 5: "Patient had critical structures contoured on the wrong CT dataset, and these were recontoured on the right dataset. Dosimetrist was called to the machine to review a CBCT and after the therapists found the carina did not match. The carina was still contoured on the wrong dataset. This was fixed and resent prior to treatment."	• Medical records incorrect • Poor human factors engineering • Slip causing physical error • Loss of attention • Expectation bias

CB, cone beam; CT, computed tomography.

as near misses and incidents increased with NMRI, whereas the proportion of events classified as "other" decreased with NMRI (Fig 1B). It is noteworthy that very few high-risk events are incidents, indicating that this group comprises mostly of events that enhanced the risk for downstream errors or had serious clinical impact but did not. Table 1 shows several example events and the causal factors that were associated with them. As expected, multiple causal factors contributed to all events, and no events were described by a single causal factor. A median of 7 causal factors were selected for each event, and this was similar for all event types: low-risk institutional (median, 6; range, 2-15), high-risk institutional (median, 7; range, 2-12), and SAFRON (median, 8; range, 2-15). The causal factor taxonomy was found to be applicable in all events. In a few cases, however, we identified 4 causal factors that were arguably not described in the AAPM taxonomy or where the categorization was not clear to all raters: linear accelerator down ($n = 3$), hardware/equipment failure ($n = 2$), failure to follow through with quality improvement intervention ($n = 1$), and misleading workflow documentation ($n = 1$).

A wide range of factors were selected for events, but certain factor categories and subcategories were common. Of the 7 top-level categories, 4 were most common: (1) "organizational management," (2) "technical," (3) "human behavior involving staff," and (4) "procedural issues" most commonly contributed to errors in all groups (Fig 2A). At the subcategory level, 6 of 19 were common: (1) "failure to detect a developing problem"; (2) "human behavior involving staff"; (3) "communication"; (4) "policies, procedures, and regulations" (further referred to as "policies"); (5) "failure to interpret the nature of a developing problem"; and (6) "equipment design and construction issues" (Fig 2B). These 6 subcategories map to the 4 most common top-level categories listed previously.

Within each of these categories there are multiple specific causal factors (see previous); Table 2 shows common specific causal factors within the 6 most common categories. These can also be seen in the examples in Table 1. We analyzed the data to determine whether there were differences in specific causal factors between the 3 types of events (eg, low-/high-risk institutional events and SAFRON events) as described in the following subsections.

Causal factors contributing more to SAFRON than institutional events

SAFRON events included more contributions from human behavior, failure to detect a problem, and policies compared with the low- and high-risk institutional groups (Table 2). The most common specific causal factor in the human behavior category (and overall) was a “slip causing physical error,” affecting 94% of SAFRON events and one-half of institutional events (Table 2, $P < .0001$). A “slip causing physical error” is defined as “a failure in performance of a highly developed skill as intended or maintained.”¹ For example, a therapist accidentally scheduling a patient for daily treatment when the prescription calls for treatment twice weekly is considered a “slip causing physical error” (Table 1, example 1). “Poor judgment” and “negligence” were rarer, but were

significantly more frequent in the SAFRON than institutional databases (Table 2, $P = .0001$). In the “failure to detect a problem” category, “loss of attention” was the most common specific causal factor, which affected 78% of SAFRON events but only 33% and 35% of low- and high-risk institutional events, respectively (Table 2, $P < .0001$). As an example, the failure to detect an incorrect setup before treating a patient is may be partially driven by loss of attention (Table 1, example 2). In the “policies” category, an inadequate policy was the most common specific causal factor, affecting 32%, 18%, and 16% of SAFRON, high-risk institutional, and low-risk institutional events, respectively (Table 2, $P = .01$). This specific causal factor was often scored when event reports described a change to a current policy or procedure as a preventive action for future events (Table 1, example 2).

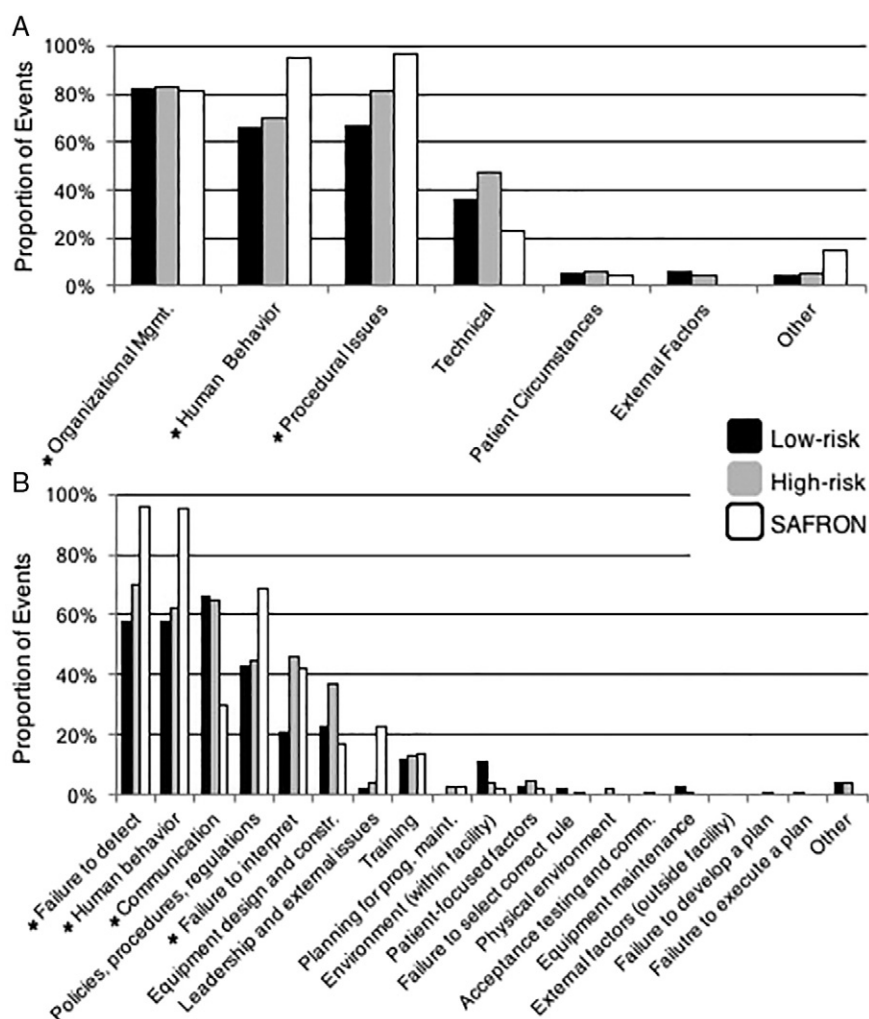


Figure 2 (A) Proportion of events in the low-risk (black), high-risk (gray), and SAFRON (white) groups attributed to each of the 7 top-level causal factor categories. (B) Proportion of events in the low-risk (black), high-risk (gray), and SAFRON (white) groups attributed to each of the 19 causal factor subcategories included in the taxonomy. Each subcategory (B) maps to 1 of the 7 top-level categories (A). “Human behavior involving staff” and “Other” have no subcategories, so they are listed as both top-level and sub-categories. *Categories or subcategories involving human factors. Abbreviation as in Fig 1.

Table 2 The most frequent specific factors within the 6 most common causal factor categories

Causal factor	Low risk (%)	High risk (%)	SAFRON (%)	P value
<i>Human behavior</i>				
Slip causing physical error	52.0	55.0	94.0	<.0001
Poor judgment	4.0	8.0	22.0	.0001
Negligence	3.0	1.0	9.0	.01
<i>Failure to detect a developing problem</i>				
Loss of attention	35.0	33.0	78.0	<.0001
Lack of information	30.0	41.0	29.0	.14
<i>Policies, procedures, and regulations</i>				
Policy inadequate	16.0	18.0	32.0	.01
Relevant policy nonexistent	16.0	15.0	24.0	.20
Policy not followed	14.0	12.0	21.0	.18
<i>Communication</i>				
Poor or incomplete	47.0	41.0	16.0	<.0001
Inadequate communication patterns	10.0	3.0	0.0	<.01
Medical records incomplete/absent	32.0	44.0	21.0	<.01
<i>Equipment design and construction</i>				
Poor human factors engineering	13.0	33.0	6.0	<.0001
Software operation failure	11.0	5.0	3.0	.05
<i>Failure to interpret the nature of a developing problem</i>				
Expectation bias	7.0	28.0	16.0	<.001
Incorrect information	2.0	11.0	16.0	<.01
Inadequate search	6.0	5.0	12.0	.13
Missing information	6.0	12.0	6.0	.20

Causal factors contributing more to institutional than SAFRON events

In contrast to SAFRON events, the most common categories in the low- and high-risk institutional events were “communication” and “equipment design and construction” (Table 2). In the “communication” category, the most common specific causal factor was “poor, unclear, incomplete, or missing communication,” which contributed to more than one-third of low- and high-risk institutional events but only 16% of SAFRON events (Table 2, $P < .0001$). Table 1 shows examples of events in which poor or incomplete information contributed to an event (examples 3 and 4). “Poor HFE” was the most common factor in the “equipment design and construction” category, and was significantly more common in the high-risk institutional dataset (33%) than in the low-risk (13%) or SAFRON (6%) datasets (Table 2, $P < .0001$). HFE is concerned with the design of tools and systems that takes into account human limitations and characteristics, with the goal of safe and effective human use.²⁴ Table 1 exhibits an example of 1 such event (example 5), a case in which a physician working with multiple CT planning scans (eg, 4-dimensional, average, maximum intensity projection) contoured the critical structures on the wrong scan. The dosimetrist redrew most of the contours on the correct scan, but sent the wrong carina contour to the treatment machine

for cone beam CT image guidance and alignment. Although the treatment team identified and addressed the problem before treatment, this patient could have been set up to an incorrect location based on the carina contour. Clearly the human-computer interface is at work in this case, and one could envision an HFE solution that could help prevent this problem, where the contouring and treatment planning software could recognize and alert providers of potential errors when contours exist across multiple scans.

The example (Table 1, example 5) also demonstrates “expectation bias,” which was the most common specific causal factor in the “failure to interpret the nature of a developing problem” category (Table 2, $P < .001$). In this example, the physician and dosimetrist were working in an established workflow and the software environment that led them to expect that all relevant contours were delineated on the correct planning scan. Note expectation bias and incorrect information were both more common in the high-risk and SAFRON groups compared with the low-risk group (Table 2, $P < .01$).

Discussion

This study examines the applicability of a comprehensive taxonomy of causal factors of error proposed in a

consensus report from the AAPM¹ and used in the RO-ILS. The taxonomy could identify causes for all events analyzed here. There were a few events where contributing factors were not listed or clearly mapped to the published taxonomy, most commonly machine downtime, but these have been addressed in the revised causal factor taxonomy used in RO-ILS (after October 2016). Because the data support the idea that errors in RO have multiple contributing factors,^{1,25,26} the taxonomy can be used as a standardized framework to facilitate identification of contributing factors during root cause analysis.

Although there are some differences between the types of events (eg, near-miss vs incident vs “other”), this study found that, in general, the causal factors that frequently contribute to events are similar (eg, communication, human behavior). This finding suggests that identification of contributing factors that lead to unsafe conditions and near-miss events might lead to process improvements that reduce the risk of more serious safety incidents that reach patients. This is important because unsafe conditions, potential process improvements, and near-miss events are much more commonly identified in high-volume institutional ILS systems than incidents that reach patients.¹⁸

Regarding specific causal factors, this study found that human behaviors and communication frequently contribute to RO errors. This is consistent with other studies,^{16,26-30} but the use of the comprehensive taxonomy in this study reveals additional information about how human behavior and communication contribute to errors in RO. For example, 1 small study analyzed events from the Radiation Oncology Safety Information System database and found skill-based errors (roughly corresponds to “physical slips”) contributed to all events, and attention failures (roughly corresponds to “loss of attention”) contributed to 13 of 25 adverse and 7 of 7 near-miss events.²⁸ The high-volume analysis presented here additionally reveals that negligence or poor judgment are exceedingly rare “human behaviors” driving errors compared with slips. Additionally, “lack of information” was found to be as frequent a contributor to error as “loss of attention” in the low- and high-risk institutional events. Likewise, International Atomic Energy Agency report 17 details numerous communication-related sources of RO errors, such as mistakes transferring information or the lack of available written information.²¹ This study expands on that data by demonstrating that some types of communication errors are very common (eg, poor communication, incorrect medical record), whereas others are more rare (eg, incorrect communication patterns). An even more detailed study of communication errors in RO would be useful in identifying targets for improvement in this area.

One specific causal factor that emerged in this study was “poor HFE.” HFE is the study of the “sociotechnical” aspects of the environment including the interactions between humans and technology,²⁴ and this was the only

specific causal factor to contribute more frequently to high-risk versus low-risk events in this study. Poor HFE design increases the mental effort needed for providers to perform tasks effectively³¹ and likely contributes to the most common “human errors” such as those discussed previously. Addressing poor HFE design in RO software and hardware will be difficult and will require close collaboration with vendors of RO equipment.³¹ Nevertheless, such an effort would be in line with a call for an engineering and health care partnership to improve the safety and efficacy of health care.³² Improvement of HFE is also a more reliable intervention for reducing the “human behavior” causes of errors described previously compared with policy changes or provider training.³³

This study is limited in a few important ways. First, the near miss events were almost exclusively limited to 1 institution and therefore the frequencies of causal factors may not be generalizable to other clinics. Second, this was a retrospective analysis of event reports, so scoring and interpretation of causal factors in the taxonomy is subjective and limited by the data available to the raters. A detailed root cause analysis was not completed on any SAFRON reports and only a small minority of institutional reports; therefore, causal factors may have been missed. Additionally, the raters had knowledge of the processes and practices for institutional events, but did not have access to this information for the clinics submitting SAFRON events. This could have biased the comparison of causal factors contributing to institutional versus SAFRON events. Further, people reporting and analyzing events may have biases about the causes of events, so causal factors were attributed to events by consensus to minimize this bias. A similar strategy should be used when reviewing events for potential contributing factors in radiation oncology clinics.

This study also only investigated events related to external beam radiation therapy, and did not include brachytherapy. In practice, causal factor analysis from ILS data should be combined with other tools such as failure mode and effects analysis, because these have been shown to reveal complementary information about risk in RO.³⁴ Because of these limitations, these findings should be considered hypothesis-generating.

Conclusions

The taxonomy used in this study was found to be applicable to all events. We believe it will be useful in performing root cause analyses by providing a framework for systematic identification of potential factors leading to errors in RO. As expected, communication and human behaviors were leading contributors to all types of errors, and this study suggests that specific types of communication errors (eg, errors in the medical record) and human

behaviors (eg, “physical slips”) contribute to errors more commonly than others. Poor HFE uniquely contributes to high-risk versus low-risk events, and commercial vendors must work with providers to improve human-technology interactions.

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